

Arena Play - Project Delivery Documentation

Scalable Live Sports Streaming Solution

Delivered by: Ifeanyi Ugwu, Senior Technical Architect

Project Duration: 6 Weeks

Delivery Date: July 30, 2025

Platform: arenaplay.cl

Executive Summary

Arena Play has been successfully delivered as a production-grade live sports streaming platform. The solution provides real-time streaming capabilities, comprehensive content management, and robust administrative controls, designed specifically for sports event broadcasting.

Key Deliverables

- **Live Streaming Infrastructure** with zero-downtime deployment capabilities
- **Content Management System** with automated cleanup and optimization
- **Administrative Dashboard** for complete platform oversight
- **Scalable Architecture** ready for payment integration and future enhancements
- **Production Monitoring** with comprehensive logging systems

Technical Achievement Highlights

- **Zero-Downtime Deployment Pipeline** using green-blue deployment strategy
 - **Efficient Resource Management** with automated image cleanup and storage optimization
 - **Enterprise-Grade Monitoring** with structured logging and rotation
 - **Court-Based Stream Management** preventing concurrent streams per venue
-

Technical Architecture

Core Technology Stack

- **Backend:** Node.js with Express framework
- **Frontend:** React/Next.js with TypeScript
- **Database:** MongoDB for all application data
- **Video Processing:** Mux for live streaming and video management
- **Process Management:** PM2 with auto-restart configuration

- **Storage:** VPS-based image and asset management

Infrastructure Specifications

- **Server:** Hostinger VPS
 - 16GB RAM
 - 4 CPU Cores
 - 200GB Storage
- **Deployment:** PM2 single instance with auto-restart
- **Monitoring:** Winston with daily log rotation

System Features

Live Streaming Management

- **Stream Lifecycle Control:** Camera activation tied to stream status
- **Court-Based Restrictions:** Single active stream per court venue
- **Real-Time Processing:** Direct integration with Mux streaming API
- **Automatic Cleanup:** 14-day retention for unsubscribed users

Content Management

- **Image Storage:** Profile photos, banners, and court images on VPS
- **Automated Cleanup:** Daily cron job for unused assets
- **Storage Optimization:** Efficient space utilization with cleanup cycles
- **Metadata Management:** MongoDB-based content relationships

Administrative Controls

- **Configuration Management:** Real-time platform settings adjustment
- **Analytics Dashboard:** Stream metrics and usage statistics
- **User Management:** Subscription and access control
- **System Monitoring:** Application health and performance metrics

Operations Manual

Administrative Access

Admin Panel URL: <https://arenaplay.cl/admin>

Dashboard Features

- **Statistics Overview:** Platform usage and performance metrics
- **Stream Management:** Active streams and court assignments
- **Configuration Controls:** System-wide settings and parameters
- **Content Management:** Asset organization and controls

Monitoring and Maintenance

Automated Systems

- **Log Management:** Winston daily rotation with structured logging
- **Asset Cleanup:** 24-hour automated cleanup cycle
- **Stream Cleanup:** Mux retention management for unsubscribed users
- **Health Checks:** PM2 automatic restart on failures

Manual Monitoring

- **Application Logs:** Available through VPS access
 - **Database Health:** MongoDB connection and performance monitoring
 - **Storage Usage:** VPS disk space and asset storage tracking
 - **Stream Performance:** Mux dashboard metrics and analytics
-

Credentials & Access Management

Access Credentials Provided

- **MongoDB Database:** Connection strings and authentication credentials
- **Mux Video Platform:** API keys and dashboard access
- **VPS Server Access:** Administrative credentials
- **Admin Panel:** Administrative account

Security Recommendations

- **Credential Rotation:** Quarterly password updates recommended
 - **Access Auditing:** Regular review of user permissions for provided accounts
 - **Backup Security:** Secure storage of credential backups
-

Payment Integration Readiness

The platform architecture is designed for seamless payment provider integration:

Integration Points

- **Subscription Management:** User account upgrade/downgrade workflows
- **Access Control:** Subscription-based feature gating
- **Billing Cycles:** Automated renewal and expiration handling

Supported Integration Types

- **Stripe:** Ready for implementation

- **PayPal:** Compatible architecture
 - **Regional Providers:** Flexible payment gateway support
 - **Custom Solutions:** Adaptable to specific requirements
-

Support & Maintenance

Complimentary Support Period

Duration: 7 days from delivery date

Coverage: Bug fixes, configuration assistance, and operational guidance

Response Time Commitments

- **Critical/Production Issues:** 24 hours
- **General Support Requests:** 72 hours
- **Enhancement Discussions:** 96 hours

Ongoing Support Options

Hourly Consulting

- **Technical Support:** System troubleshooting and optimization
- **Feature Development:** New functionality implementation
- **Performance Tuning:** System optimization and scaling
- **Integration Assistance:** Third-party service integration

Monthly Retainer Package

- **Proactive Monitoring:** Regular system health checks
- **Priority Support:** Reduced response times
- **Maintenance Updates:** Security patches and system updates
- **Strategic Consultation:** Growth planning and architecture advice

Contact Information

Technical Architect: Ifeanyi Ugwu

Support Channel: valentineifeanyiugwu@outlook.com

Business Hours: Monday - Friday, 9 AM - 6 PM WAT

Future Enhancement Roadmap

Performance Optimizations

- **Redis Integration:** Database query caching for improved response times

- **PM2 Cluster Mode:** Load balancing for increased concurrent capacity
- **CDN Implementation:** Global content delivery optimization
- **Database Indexing:** Query performance optimization

Scalability Improvements

- **Containerization:** Docker implementation for deployment flexibility
- **Microservices Architecture:** Service separation for independent scaling
- **Load Balancing:** Multi-server deployment capabilities
- **Auto-Scaling:** Dynamic resource allocation based on demand

Advanced Features

- **Real-Time Analytics:** Enhanced streaming metrics and insights
- **Advanced Monitoring:** Comprehensive system observability
- **API Rate Limiting:** Enhanced security and resource protection
- **Multi-Region Support:** Geographic distribution capabilities

Monitoring Enhancements

- **Application Performance Monitoring (APM):** Detailed performance insights
- **Error Tracking:** Advanced debugging and issue resolution
- **Custom Dashboards:** Business-specific metrics visualization
- **Alerting Systems:** Proactive issue notification

Project Completion Summary

Arena Play has been delivered as a robust, production-ready streaming platform that meets all specified requirements. The implementation focuses on reliability, scalability, and maintainability, providing a solid foundation for growth and future enhancements.

Technical Excellence Delivered

- **Zero-Downtime Deployment:** Ensures continuous service availability
- **Resource Optimization:** Efficient storage and bandwidth utilization
- **Comprehensive Monitoring:** Full system observability and logging
- **Security Best Practices:** Secure credential management and access control

Business Value Delivered

- **Immediate Deployment:** Platform ready for production use
- **Scalable Architecture:** Supports business growth and expansion
- **Payment Ready:** Streamlined path to monetization
- **Professional Support:** Ongoing technical partnership available

The platform is now ready for production use and positioned for successful scaling as your user base grows.